

Request for Information
Air Traffic Systems Branch
Multifunction Tactical Radar System (MTRS)
8 April 2026

This is a Request for Information (RFI) only, which as part of market research, is being issued in accordance with (IAW) Revolutionary Federal Acquisition Regulation Overhaul (RFO) part 10. This is not a solicitation/Request for Proposal, a Request for Quotation, an Invitation for Bids, or a solicitation and no contract shall be resultant from this synopsis, **Aerospace Management Systems Division (AFLCMC/ESA) will not pay respondents for information provided in response to this RFI.** The Air Force Life Cycle Management Center's (AFLCMC) Electronic Systems Directorate, Aerospace Management Systems Division (ESA), Air Traffic System Branch (ESAA) Program Management Office (PMO), as a Technology Project on the investment master list (IML), located at Hanscom Air Force Base (AFB), Massachusetts is requesting information from industry to assist in the planning for the future acquisition of a Multifunction Tactical Radar System (MTRS).

Phase 1 of the Other Transaction Agreement (OTA) for MTRS, which was awarded on 23 February 2024, was completed during February 2025 and currently Phase II is underway with planning to end FY Q4 2026. This RFI is to continue on-going market research to understand and keep in touch with industry base. MTRS is being developed to execute four missions, which are: (1) ATC surveillance, (2) Battle Management Command and Control (BMC2), (3) Unmanned Aircraft Systems (UAS) and (4) Weather Sensing.

The Government encourages industry partners to supplement their formal response by identifying any existing system, subsystems and/or major components (e.g., AESA panels, T/R modules, resource manager) that are representative of their proposed design and meet MTRS packaging and transport requirements (e.g., fit, form and function). Please provide this information within five working days of receiving this RFI, so we can arrange further discussions to evaluate the components' applicability and determine the next steps.

The Government is looking for responses to the below questions for potential complete or partial system solutions or enabling technologies. Please respond to the best of your ability to as many questions as possible. Solutions should be actionable within the next year (FY2027). Also, address your ability to manufacture and deliver initial pre-production unit(s) (PPUs) in 29 months after award.

DESCRIPTION: The Department of War (DoW) transformation vision relies heavily on more expeditionary and agile joint forces which in turn places increased emphasis on aviation assets for deployment, employment, sustainment, and redeployment. To support this vision, the Air Force requires the ability to launch and recover sorties from an agile network of airbases with varied capabilities in all weather conditions while also providing localized situational awareness for the protection of those airbases. Future ATC systems adaptability is critical to meet short and enduring missions. The MTRS shall detect airborne targets and produce data for varied systems to enable timely and relevant BMC2 actions (Air Traffic and Airspace Control) and defensive decisions against threats. The MTRS shall provide real-time Doppler weather information including Next Generation Weather Radar (NEXRAD) Level II base products (reflectivity, mean radial velocity, and spectrum width) that are crucial for meteorological analysis and weather forecasting. The MTRS sensors shall support all four (4) mission functions individually and simultaneously. The MTRS shall provide the scalable sensor coverage for rapid decision-making and improve situational awareness needed to support Adaptive Operations in Contested Environments (AOiCE). To support the emerging AOiCE Concept of Operations (CONOPs), the MTRS shall be capable of withstanding world-wide deployable environmental conditions and the unique rigors of repeated world-wide deployable cycles in a reliable and sustainable manner.

EXPECTED MINIMUM PERFORMANCE: The Air Force requires the ability to launch and recover sorties from an agile network of airbases with varied capabilities in all weather conditions while also providing localized situational awareness for the protection of those airbases. In order to execute the scheme of maneuver inherent with Agile Combat Employment (ACE) while contributing to the theater

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sensing grid in support of Advanced Battle Management System (ABMS), lightweight multifunctional sensors are required. These sensors must enable effective terminal ATC, airspace warning, weather processing, BMC2 and response to threats operations while supporting the maneuverability and survivability required. As a multifunction platform, MTRS is intended to reduce the volume of logistical support for inter/intra theater airlift, supply chains, and support personnel and result in a reduction of electromagnetic interference from disconnected/independent systems trying to work simultaneously.

The MTRS must be highly configurable and capable of performing the defined mission functions simultaneously to the greatest extent:

- **ATC Surveillance:** Enabling aircraft identification, separation, and sequencing in a terminal environment in both instrument flight and visual flight rule (IFR/VFR) operations
- **BMC2 Surveillance:** Enabling the detection, identification, classification, and tracking of hostile undeclared manned and unmanned aircraft
- **Unmanned Aircraft Systems (UAS) Surveillance:** Enabling the detection of Group 1 (0-20 lbs.), Group 2 (21-55 lbs.) and Group 3 (<1,320 lbs.)
- **Weather Sensing:** Providing real time Doppler weather information including reflectivity, radial velocity, and spectrum width

1. Key Radar Performance Requirements:

ATC Surveillance (Primary Radar) Performance Requirements

Parameter	Threshold	Objective
Detection Range (nautical mile, NM)	0.5-60	0.5- 60
Azimuth Coverage (deg)	0-360	0-360
Elevation Coverage (deg)	0-30	0-30
Altitude Coverage (ft. AGL)	100-21,000	100-24,000
Target RCS (m ²)	1.0 (to 60 NM)	1.0 (to 60 NM)
Target Capacity	250	450
Target Radial Velocity (knots)	0 to 700 (clear) 20-700 (elevated clutter environments)	0 to 700 (clear) 20-700 (elevated clutter environments)
Probability of Detection (%) (Cumulative probability of detection over the update period for a 1 m ² target at maximum range)	80	90
Probability of False Alarm (Reference value to support overall detection performance and may be varied provided the design can deliver acceptable false plot and track report performance.)	10 ⁻⁶	10 ⁻⁶

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Parameter	Threshold	Objective
Range accuracy (ft RMS)	275	250
Azimuth Accuracy (deg RMS)	0.25	0.16
Update rate (sec)	5.0	4.0
Latency (sec)	0.8	0.8
Weather Detection and Reporting (Detect, categorize, and report precipitation within the six reflectivity levels defined by the National Weather Service (NWS).)	Range: 18 to 57 dBZ Levels (dBZ): Level 1: (18-29 dBZ) Level 2: (30-40 dBZ) Level 3: (1-45 dBZ) Level 4: (46-50 dBZ) Level 5: (51-56 dBZ) Level 6: (>57 dBZ)	Range: 18 to 57 dBZ Levels (dBZ): Level 1: (18-29 dBZ) Level 2: (30-40 dBZ) Level 3: (1-45 dBZ) Level 4: (46-50 dBZ) Level 5: (51-56 dBZ) Level 6: (>57 dBZ)
Weather Update Rate (sec)	60	30

ATC Secondary Surveillance Radar Performance Requirements

Parameter	Threshold	Objective
Modes	1, 2, 3A, C, S, 5 level 1	1, 2, 3A, C, S, 5 level 1 and 2
Passive ADS-B (NM)	100	120
Range (NM)	100	120
Azimuth Coverage (deg)	0-360	0-360
Elevation Coverage (deg)	0-30	0-30
Altitude Coverage (ft. AGL)	60,000	100,000
Target Capacity	250	450
Target Velocity (knots)	25-1290	0-2580
Probability of Detection (%)	97	97
Range Accuracy (ft RMS)	190	175
Azimuth Accuracy (deg RMS)	0.25	0.16
Update rate (sec)	5.0	4.0
Latency (sec)	0.8	0.8

NOTE: Secondary radar will also support BMC2 and UAS surveillance functions

BMC2 Surveillance Performance Requirements

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Parameter	Threshold	Objective
Range (NM)	4.0-100	0.5-120
Azimuth Coverage (deg)	0-360	0-360
Elevation Coverage (deg)	0-30	0-30
Altitude Coverage (ft. AGL)	60,000	100,000
Target RCS (m ²)	0.1 at 60 NM 1.0 100 NM	0.1 at 80 NM 1.0 at 120 NM
Target Capacity	250	500
Target Radial Velocity (knots)	up to 1290 (clear) 20 to 1290 (other clutter environments)	0 to 2580 (clear) 20 to 2580 (other clutter environments)
Probability of Detection (%)	90	95
Probability of False Alarm (Reference value to support overall detection performance and may be varied provided the design can deliver acceptable false plot and track report performance.)	10 ⁻⁶	10 ⁻⁶
Elevation Accuracy (deg RMS)	0.25	0.09
Azimuth Accuracy (deg RMS)	0.25	0.17
Range Accuracy (ft RMS)	100	100
Update Rate (sec)	10	5
Latency (sec)	0.8	0.8

UAS Surveillance Performance Requirements

Parameter	Threshold	Objective
Range (NM)	0.5-5	0.5-10
Azimuth Coverage (deg)	0-360	0-360
Elevation Coverage (deg)	80 (-10 deg to +70 deg)	80 (-10 deg to +70 deg)
Altitude Coverage (AGL)	See Table immediately below for details	
Target RCS (m ²)	0.01 to 0.1	0.01 to 0.1
Target Capacity	20	40
Target Radial Velocity (knots)	5-250 (clear)	0-300 (clear) 0-300

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Parameter	Threshold	Objective
	5-250 (other clutter environments)	(other clutter environments)
Probability of Detection (%)	95	99
Probability of False Alarm <i>(Reference value to support overall detection performance and may be varied provided the design can deliver acceptable false plot and track report performance.)</i>	10^{-6}	10^{-6}
Elevation Accuracy (deg RMS)	1.0	0.5
Azimuth Accuracy (deg RMS)	1.0	0.5
Range Accuracy (ft RMS)	100	30
Update Rate (sec)	1.0	0.25 short range <1 long range
Latency (sec)	0.8	0.8

UAS Group:

Group	Max Weight (lb)	Operating Altitude (ft)	Speed (kts)
1	0-20	<1,200 AGL	<100
2	21-55	<3,500 AGL	<250
3	<1,320	<18,000 AGL	<250

Weather Sensing Performance Requirements

Parameter	Threshold	Objective
Max Reflectivity Detection Range (NM)	124	180
Min Reflectivity Detection Range (NM)	5	0.5
Doppler Detection Range (NM)	50	50
Altitude Coverage (ft AGL)	0-70,000	0-70,000
Azimuth Coverage (deg)	0-360	0-360
Elevation Coverage (deg)	32 (-2 deg to +30 deg)	92 (-2 deg to +90 deg)
Minimum Precipitation Detection (throughout coverage volume)	1 inch per hour	1 inch per hour
Precipitation Levels	16 from 0 dBz	16 from 0 dBz
Mean Radial Wind Velocities Detection (throughout coverage volume)	up to 50 knots	up to 50 knots

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Parameter	Threshold	Objective
Update Rate (sec)	240	120
Weather Detection and Reporting (Detect, measure, and report meteorological phenomena as defined in NEXRAD ICDs)	Level II (Base Products)	Level III (Derived Products)

Definitions:

- “Elevation Coverage” requirements shall be referenced to the horizontal plane of the radar site horizon.
 - “Target RCS” defines the smallest target size that the MTRS system is required to detect and report. Additionally, the MTRS system shall have the capability to detect and report on larger targets (i.e., aircraft or UAS) as required for each mission.
 - Regarding “Target Capacity”, the MTRS system should be capable of detecting and reporting plot or track reports consisting of real target returns, uniformly or non-uniformly distributed, per scan, while operating in clear or elevated clutter environments.
 - Accuracy performance (range, azimuth, and elevation angle) shall be based on measured position data (i.e., extracted from a plot report) not track updates (i.e., smoothed track position).
 - “Latency” is defined as the time from when target reply signals were present at the radar antenna to the time the plot report is available for transmission on the external interface.
2. **Transport:** The MTRS, when configured for transport, shall be capable of being transported via military cargo aircraft, ground transport by semi-trailer flatbed trucks, military trailers and trucks, railcar, military sea-going vessels and sling load transport via military helicopters (CH-47).

MTRS is expected to be transported via helicopter sling load, fixed-wing aircraft, ocean-going ship, rail and ground vehicle(s) and shall withstand these transports without damage. Some of these transport conditions include:

- a. Rapid decompression (altitude change from 8,000 ft to 45,900 ft in less than 15 seconds)
- b. Military transport shock and vibration, including Type IV Road mobility per SAE AS8090 and rail impact
- c. Sea level to 45,900-foot altitude
- d. Single Apex Lift
- e. Helicopter Sling Load

The MTRS, when configured for transport, shall be capable of roll-on/roll-off (RO/RO) from Large Medium Speed Roll-on/roll-off (LMSR) ships and no more than two C-130J-30 aircraft. When configured for transport, the MTRS’ footprint, including generator sets, and all other needed ancillary equipment, shall be no larger than ten 463L pallet positions (excluding the ramp position); i.e., seven positions in the first aircraft and three positions in the second aircraft.

The overall transport weight, including the weight of four personnel and their associated gear (total of 1,400 lbs.), shall not exceed the C-130J-30 peacetime payload weight limit of 34,000 pounds. IAW DTR 4500.9-R, Part iii, Appendix V and maintain aircraft walkway clearances IAW MIL-STD-1791.

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The system shall be capable of single apex top lift for crane loading. The system shall tolerate vibration and shock induced loading during all transport modes identified herein, including rail impact and Type IV ground mobility as defined in SAE AS8090.

3. **Setup and Configuration:** Setup time for MTRS using four technicians shall take no more than four hours once the equipment is emplaced.
4. **Power:** The MTRS shall operate using external power including power transmitted at nominal frequencies of 50 and 60 Hz and at voltages throughout the nominal ranges of 100 to 130 VAC and 200 to 260 VAC.
5. **Environmental Conditions:** The MTRS shall be operable in all parts of the world and in most environmental conditions. Some of the more extreme operating conditions include:
 - Temperature Range: -54°C to +71°C
 - 5 to 100% humidity
 - Solar radiation effects
 - Blowing sand and dust
 - Salt atmosphere
 - Rain/snow/ice
 - Fungus
 - Sea level to 13,100-foot altitude
 - 50 mile per hour winds with 71 mile per hour gust (with and without ½-in ice)
6. **Operational Reliability and Maintainability:** The MTRS shall provide the following Reliability and Maintainability performance:
 - a. Mean-Time-To-Repair (MTTR): 30 minutes or less
 - b. Maximum Time to Repair (MaxTTR): ≤1.5 hours
 - c. Mean-Time-Between Critical Failure (MTBCF): ≥3000 hours

REQUEST: Request vendors provide potential MTRS solutions which vendors believe fit the above criteria. The government would prefer a Commercial-Off-The-Shelf (COTS) solution. However, market research to date has shown that some level of development may be required.

Please provide responses to as many of the tasks listed below as possible. Address your capabilities to accomplish part or all the below tasks and provide highlights of successful past experience on similar government contracts within the past five (5) years.

1. Provide a **BRIEF** description of your company's expertise involving mobile or deployable radar systems.
2. Describe your capability to provide a MTRS solution that meets the expected minimum performance described above.
3. Provide a description of your existing systems or equipment that can be used to provide the capabilities described above.
 - a. What, if any, modifications of your existing equipment (including software) will be necessary to meet MTRS performance requirements. If development is required, please provide a description of the level of developmental effort, including technical description

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for each major subcomponent as well as risks and ways to mitigate the risks. Pictures and illustrations of the proposed MTRS and its major subcomponents would be appreciated.

- b. Provide your assessment of the current Technology Readiness Level (TRL) and Manufacturing Readiness Level (MRL) of the MTRS your company would provide per the DoW TRL and MRL Deskbook, including substantiation of the levels chosen with specifics regarding supporting information such as test reports.
- c. Explain any limitations and challenges for meeting the MTRS requirements. Provide details regarding any expected minimum performance that your recommended MTRS solution cannot meet. Identify any MTRS requirements that appear to be unachievable or likely cost drivers. Present details on trade space issues and opportunities.
- d. Describe your system's surveillance capability to detect and report airborne objects throughout the coverage volumes (range, azimuth, elevation, and altitude) defined for each mission (ATC, BMC2, UAS, and Weather Sensing) simultaneously to the greatest extent.
- e. Describe the maximum number of targets that your system can simultaneously detect and any throughput limitations.
- f. Describe the minimum and maximum aircraft velocities that your system can detect. Describe the challenges to detect the slowest and fastest aircraft velocities, and other limitations. Explain your system's clutter processing capabilities in worldwide operating environments such as distributed surface clutter (e.g., lowlands, wooded hills, mountains, urban areas), fixed or moving discrete clutter (e.g., towers, vehicles), sea clutter, rain. Include discussion for related environmental considerations such as Anomalous Propagation (AP), ducting, angels (e.g., birds), and wind turbines.
- g. Describe your system's capability to detect, locate, and mitigate electronic interference. Describe your systems available data products, formats and interfaces as related to DoW interoperability expectations and architectures. Identify the key interfaces to your system. Are Interface Control Documents (ICD), or other applicable documentation, available for these interfaces?
- h. Describe your system's capability to provide plot and track reports at the external interface to support integration into the respective Command and Control (C2) systems and operational requirements of the applicable mission functions.
 - i. Plot Report: A plot is a single observation from a detected target. It may consist of a few detections in neighboring ranges. However, through processes of clustering and centroiding, one single observation is associated with the neighboring detections. This process is called Plot Extraction. Plot reports may be referred to as plot reports and uncorrelated plot reports.
 - ii. Track report: A track is a series of plots belonging to one (moving) target which are observed and formed in consecutive radar scans. Track reports may be referred to as correlated plot reports. Note: Estimated track updates (i.e., coasted track reports) shall not be available at the external interface unless those messages are identified as unassociated track messages.
 - iii. The plot or track reports available at the external interface originate from associated plot reports and contain the associated position information.
 - iv. Plot and/or track reports availability at the external interface selectable by mission function.
- i. Describe your systems ability to meet the ATC azimuth accuracy requirements (Threshold) designed to enable terminal approach control radar separation services of 3-NM when inside of 40-NM from the antenna and 5-NM when greater than 40-NM from the antenna as defined in Federal Aviation Administration (FAA) Order JO 7110.65, ICAO Doc 9924, and EUROCONTROL Specification for ATM Surveillance System Performance (Vol 2).

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- j. Describe the transportable design of your proposed MTRS solution. Discuss any limitations (e.g., ACE deployment).
 - k. Discuss the scalability of your existing equipment to deliver a sensor that supports an alternative acquisition strategy. Considering the evolving operational environment, the government may require a sensor that supports a subset of the MTRS mission functions such as an ATC and UAS only sensor. Provide an overview of the expected design, fabrication, schedule, and cost factors, along with potential challenges, to accommodate an alternative acquisition strategy.
 - l. BMC2 and UAS Target Identification: Responder shall describe their target classifier implementation approach and expected precision (e.g., probability of correct classification) as well as recommended training and validation strategies.
4. Explain if there are any other advantages or disadvantages to your recommended MTRS solution with an emphasis in the following areas:
 - a. Operating in a contested environment (electronic protection, degraded GPS, etc.)
 - b. Agile deployment worldwide complemented with self-optimization and adaptive operations
 - c. AI/ML applications to enhance surveillance capabilities.
 5. Has your system been verified to meet an open architecture framework? If your solution is compliant to an open architecture framework, please describe the standard and the certification process as applicable.
 6. What hardware and software open standards does your system use or comply with? Please describe any verification activities that may have been completed for proposed components.
 7. Provide rough order magnitude (ROM) cost and schedule for the following as the ROM will be used as a part of a feasibility analysis to determine whether future efforts can be performed within the program budget using the table below:

Rough Order Magnitude Cost Estimate			
Base Year			
Suggested LRIP Qty			
Category	Labor	Materials	ODCs
a. Prototype Design			
b. Prototype Fabrication			
c. System Level Testing			

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d. Environmental Testing			
e. Annual Maintenance Cost per System			
DLR Repairs			
Consumables			
Software			
Operators			
Maintainers			
f. Cost Drivers			
g. Schedule Drivers			
h. Concerns			

8. Provide any additional information you consider relevant to providing a MTRS solution.
9. Please provide your capability statement and include your NAICS and Small Business Certifications. Additionally, include if you are a Non-Traditional Contractor, and if you are on any Multiple Award IDIQ contracts or part of any OT Consortium.

Submissions should be delivered via DoW's Safe Access File Exchange (SAFE) website at <https://safe.apps.mil/>. Since unauthenticated users (Non-DoW Personal Identity Verification (PIV) certificates) must receive a drop-off request from a DoW user, please notify the Contracting Officer/Agreements Officer Ms. Tiffany N. Hinson (tiffany.hinson.1@us.af.mil) of your company's intent to deliver a response. You will be provided with a "Drop-off Request" which you will use to submit your files. The Drop-Off Request will be valid for 14 days. For additional guidance, please see the SAFE website at <https://safe.apps.mil/>.

Responses to this RFI are requested by close of business on 29 April 2026. All responses shall be unclassified and reviewed to ensure consideration of operational sensitivities. To the maximum extent possible, please submit nonproprietary information. Any proprietary information submitted should be identified as such and will be handled accordingly and protected from disclosure. Proprietary information will be safeguarded in accordance with the applicable Government regulations. The Government shall not be liable for damages related to proprietary information that is not properly identified. Any submissions in response to this RFI constitutes consent for that submission to be reviewed by Government personnel, Federally Funded Research and Development Center (FFRDC) Contractor employees, and Advisory & Assistance Services (A&AS) Contractor employees supporting the Air Force AFLCMC/ESA who have signed Non-Disclosure Agreements (NDA) – unless the respondent clearly objects in writing to the release of this information to FFRDC and/or A&AS Contractor employees in a cover letter accompanying the respondent's submission. All Government and DoW contractor personnel reviewing RFI responses understand their responsibility for proper use and protection from unauthorized disclosure of proprietary information. All correspondence related to this matter should be e-mailed to:

Contracting Office Address:

AFLCMC/ESAK
75 Vandenberg Drive
Hanscom Air Force Base (AFB), MA 01731-2103
United States

Primary Contracting Point of Contact:

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Tiffany Hinson
Contracting Officer
tiffany.hinson.1@us.af.mil
Phone: 1 380-456-6643 (TEAMS)

Secondary Contracting Point of Contact

Robert Sullivan
Contract Specialist
Robert.Sullivan.39@us.af.mil

Primary PM Point of Contact:

Mike Robblee
Program Manager
Michael.robblee.3@us.af.mil
Phone: 1-380-456-4925 (TEAMS)